



Association between age at first cannabis use, anxiety and depressive symptoms, and cannabis dependence by sex among U.S. young adults

Karissa Tran^{a,1}, Zerla Marie Oca^a, Milly Lopez^a, Jessica Orea^{a,2}, Margot Trogden^a, Kai Orozco^{a,3}, Aryanna Alarcon^a, Haylee DeLuca Bishop^a, Susan Stewart^b, Kim Pulvers^{a,*}

^a California State University San Marcos, San Marcos, CA, USA

^b University of California, Davis, Davis, CA, USA

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ABSTRACT

Background: This study investigates whether the association between age at first cannabis use, anxiety and depressive symptoms (modeled alone and together), and cannabis dependence varies by sex.

Method: Cross-sectional online study with 451 U.S. young adults in 2021 who used cannabis in the past 30 days.

Results: In the model with both anxiety and depression, cannabis dependence was associated with anxiety symptoms only in males ($p = .0001$) and depressive symptoms only in females ($p = .007$).

Conclusion: Modeling anxiety and depression co-occurrence provided unique insight into sex-specific associations between mental health symptoms and cannabis dependence.

1. Introduction

Anxiety and depression, which are prevalent among young adults (Alzueta et al., 2023; Goodwin et al., 2020) are associated with both earlier age at first cannabis use (Bataineh et al., 2023, 2024) and greater cannabis dependence (Lawn et al., 2022), providing a target to examine as an explanatory variable. Furthermore, it is important to understand whether the expected association between earlier age at first cannabis use, higher current anxiety and depressive symptoms, and greater current cannabis dependence operates the same for males and females.

Sex differences in anxiety and depression rates (Leach et al., 2008), as well as cannabis dependence (Jeffers et al., 2021), may create different patterns of association for males and females. Identifying these patterns could inform sex-tailored prevention and treatment approaches. Given that anxiety and depression are distinct syndromes while also commonly co-occurring (Ter Meulen et al., 2021), we will test models in which anxiety and depressive symptoms are isolated as well as modeled together.

2. Method

2.1. Participants

U.S. young adults with past 30-day cannabis use ($N = 451$) in a cross-sectional survey administered through an online panel in 2021 were the focus of this study. Stratification and panel procedures are provided in a separate article (Pulvers et al., 2024). All procedures were approved by the California State University San Marcos Institutional Review Board.

2.2. Measures

2.2.1. Age at first cannabis use

The open-ended question "At what age did you first use any form of marijuana?" was used.

2.2.2. Anxiety

The Generalized Anxiety Disorder Scale-7 (Spitzer et al., 2006) ranges from 0 to 21, with scores from 5 to 9 indicating mild anxiety, 10 to 14 indicating moderate anxiety, and 15 and higher indicating severe anxiety ($\alpha = .92$).

* Corresponding author. Department of Psychology, California State University San Marcos, 333 S. Twin Oaks Valley Rd, San Marcos, CA, 92096, USA.
E-mail address: kpulvers@csusm.edu (K. Pulvers).

¹ Karissa Tran is now at University of California, Los Angeles, Los Angeles, CA

² Jessica Orea is now at Oregon State University, Corvallis, Oregon

³ Kai Orozco is now at University of Washington, Seattle, WA.

2.2.3. Depressive symptoms

The Patient Health Questionnaire-2 (Kroenke et al., 2003) ranges from 0 to 6, with scores of 3 or higher indicating major depression ($\alpha = .78$).

2.2.4. Cannabis dependence

The Cannabis Use Disorder Identification Test-Revised (CUDIT-R) (Adamson et al., 2010) ranges from 0 to 32, and 8 or more indicate hazardous cannabis use, while scores of 12 or more indicate possible cannabis use disorder ($\alpha = .83$).

2.3. Analytic approach

Population estimates for weighting our survey sample were gathered from 2021 U.S. Census Bureau state-level demographic data. Moderated mediation (Supplemental Fig. 1) was tested using Baron and Kenny's method (Baron and Kenny, 1986) in SAS Enterprise Guide version 8.3 (SAS 9.4). All models controlled for age, education, race, and cannabis legality in the state of residence and accounted for the stratified sampling design and weighting. Statistical significance was assessed at the 0.05 level (2-sided).

3. Results

Sample characteristics are displayed in Supplementary Table 1. As shown in Table 1, cannabis dependence was significantly associated with younger age at first use in both females and males, with no significant sex interaction (Model 1). When modeled alone, anxiety symptoms were significantly associated with cannabis dependence (Model 2), with no sex interaction. Similarly, when modeled alone, depressive symptoms were significantly associated with cannabis dependence (Model 3), with no sex interaction. When anxiety and depression were modeled together (Model 4), there were significant sex differences in the associations. Cannabis dependence was significantly associated with anxiety symptoms only in males ($p = .0001$) and depressive symptoms only in females ($p = .007$). Additionally, there was a significant sex interaction for anxiety ($p = .003$) and depressive symptoms ($p = .018$).

4. Discussion

Modeling anxiety and depression together revealed unique insights

into sex-specific associations between mental health symptoms and cannabis dependence. When both anxiety and depression are present, individuals face a higher overall burden of distress (Mason et al., 2025). Elevated symptom load may elicit different responses in males and females based on unique biological or psychosocial characteristics (Zahn-Waxler et al., 2008). Co-occurrence amplified the association between anxiety and cannabis dependence for males, and depression and cannabis dependence for females. Higher rates of depression in females emerge in puberty (Kessler, 2003). It is possible that earlier exposure to cannabis affects mood neuromodulation in females (Lichenstein et al., 2022) in a way that persists into young adulthood. There is evidence that depressive symptoms precede substance use issues in females (Zilberman et al., 2003).

Sex differences in social-emotional characteristics present from early childhood in which males exhibit more activity, irritability, and impulsivity (Zahn-Waxler et al., 2008), could explain why hyperarousal (i.e., feeling on edge, worrying) vs. deficit symptoms (i.e., little pleasure, feeling down) amplified anxiety results for males. It is also possible that our measure was signaling social anxiety in our male respondents, which has been associated with cannabis dependence in males (Buckner and Zvolensky, 2014). Our cross-sectional design makes it impossible to determine the causal pathway between anxiety and depressive symptoms and cannabis dependence. Prospective designs are needed to understand how to best apply our correlational findings to interventions. Nonetheless, evidence that the relationship between mental health symptoms and cannabis dependence is bidirectional (Lydiard et al., 2023) suggests that multicomponent interventions that target anxiety and depressive symptoms along with cannabis dependence may be useful. Routine screening for anxiety and depression co-occurrence should be implemented in practice to address the historically high levels of cannabis use and dependence among young adults, which demand public health attention.

CRedit authorship contribution statement

Karissa Tran: Writing – original draft, Conceptualization. **Zerla Marie Oca:** Writing – review & editing, Resources. **Milly Lopez:** Writing – review & editing, Resources. **Jessica Orea:** Writing – review & editing, Resources. **Margot Trogden:** Writing – review & editing, Resources. **Kai Orozco:** Writing – review & editing, Resources. **Aryanna Alarcon:** Writing – review & editing, Resources. **Haylee DeLuca Bishop:** Writing – review & editing, Resources. **Susan Stewart:** Writing – original draft,

Table 1

Testing the mediating effect of anxiety and depressive symptoms and the moderating effect of sex on cannabis dependence.

DV: Cannabis Dependence	Females			Males			Int. ^b
	Coeff. ^a	SE	p	Coeff. ^a	SE	p	p
Model 1							
Age at First Use	0.6085	0.2052	0.0032	−0.4709	0.1394	0.0008	0.5783
Anxiety Symptoms	NA			NA			
Depressive Symptoms	NA			NA			
Model 2							
Age at First Use	−0.5476	0.1930	0.0048	−0.3918	0.1247	0.0018	0.5017
Anxiety Symptoms	0.2582	0.1154	0.0259	0.5099	0.0942	<0.0001	0.0914
Depressive Symptoms	NA			NA			
Model 3							
Age at First Use	−0.4364	0.1780	0.0147	−0.4109	0.1341	0.0023	0.9079
Anxiety Symptoms	NA			NA			
Depressive Symptoms	1.3842	0.4095	0.0008	0.9702	0.3754	0.0101	0.4601
Model 4							
Age at First Use	−0.4551	0.1795	0.0117	−0.3864	0.1217	0.0016	0.7486
Anxiety Symptoms	−0.0133	0.1366	0.9225	0.5749	0.1467	0.0001	0.0027
Depressive Symptoms	1.4118	0.5222	0.0072	−0.3076	0.5228	0.5566	0.0183

Notes.

^a Coefficient.

^b Interaction.

Formal analysis. **Kim Pulvers:** Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

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Declaration of competing interest

All authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.psycom.2025.100228>.

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